Refraction

antocciphysics | TEACHER KEY | CPA + MATCHING SEI

Learning target: Connect a change in wave speed to wavelength and direction.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 6, Day 2.*
5 cycles; frequency 1/3 Hz ≈ 0.3333 Hz.

02. Worked example: explain the model
Refracted angle to normal smaller in the slower medium, for ordinary positive-index media.

03. Read and interpret the model
Toward normal; refracted angle smaller.

04. Apply the physics
4 m and 2 m.

05. Explain your reasoning
70° from normal.

06. Review: connect an earlier idea *Review: Unit 5, Day 5.*
No. Equilibrium must lie between 14°C and 56°C under the assumptions. Check readings or an unmodeled input.

07. Exit check
Away from normal, for transmitted ray below critical conditions.

08. STRETCH: test the model
Normal incidence remains straight; oblique incidence bends toward normal. Speed changes in both.

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01. Warm-up: bring back an earlier idea *Review: Unit 6, Day 2.*
$\lambda = 0.3$ m; $f = 1/T \approx 20$ Hz.

02. Worked example: explain the model
Refracted angle to normal smaller in the slower medium, for ordinary positive-index media.

03. Read and interpret the model
Toward normal; refracted angle smaller.

04. Apply the physics
$\theta_2 = \sin^{-1}(1/3) = 19.5°$.

05. Explain your reasoning
70° from normal.

06. Review: connect an earlier idea *Review: Unit 5, Day 5.*
Water gains 25200 J; metal $c \approx 1050$ J/(kg·°C). Equal energy loss/gain applies under the assumptions.

07. Exit check
41.8°; total internal reflection, not an ordinary refracted ray.

08. STRETCH: test the model
$\theta_2 = \arcsin[(1.33/1.50)\sin 40°] \approx 34.7°$. Reverse critical angle = $\arcsin(1.33/1.50) \approx 62.5°$.